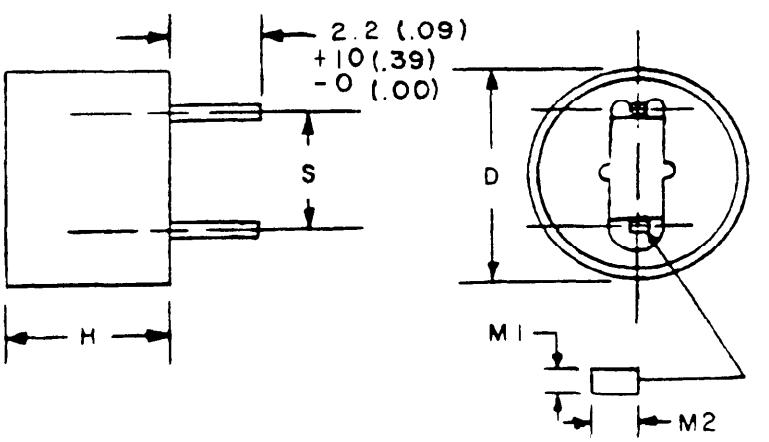
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC, DOUBLE LAYER, CARBON, (NONPOLARIZED),

STYLE DLO1 (METRIC)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and the latest issue of DOD-C-29501.



NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Inch equivalents are in parentheses.

FIGURE 1. Style DLO1 dimensions and configuration.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case material: Metal.

Leads: Solder-coated metal.

DC rated voltage: 5.5 volts.

Operating temperature: -40°C to +70°C.

DC leakage (DCL): See table I.

TABLE I. Style DL01 dimensions and characteristics.

Military part number 1/	Capacitance	Capacitance tolerance	Rated dc voltage	DC leakage (max)	R_C	ESR	D ±0.5	H (max)	S ±0.5	M1 ±0.1	M2 ±0.1	Weight typical 2/
D29501A01	Farads	Percent	Volts	µA	Ohms	Ohms	mm	mm	mm	mm	mm	Grams
B223V	0.022	-20,+80	5.5	33	1,000	50	13.0	15.0	5.1	0.4	1.2	3.0
B473V	0.047	"	"	71	1,000	40	14.5	15.0	5.1	0.4	1.2	3.5
B104V	0.10	"	"	150	100	45	14.5	24.0	5.1	0.4	1.2	7.0
B224V	0.22	"	"	330	100	25	16.5	25.0	5.1	0.4	1.2	8.0
B474V	0.47	"	"	710	100	13	21.5	25.0	7.6	0.6	1.2	14.0
B105V	1.0	"	"	1,500	10	7	28.5	25.0	10.2	0.6	1.4	24.0

1/ The complete part number will be D29501A01B223V.

2/ For information only.

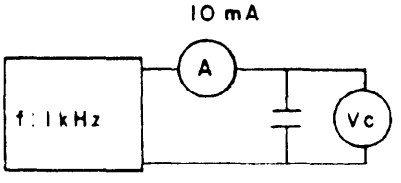
Capacitance: See table I.

Capacitance tolerance: -20, +80 percent.

ESR: See table I, and in accordance with DOD-C-29501, except circuit shall be as follows:

Circuit diagram and formula:

$$ESR = \frac{V_C}{10^{-2}} \quad (\Omega)$$



Thermal shock:

DCL: In accordance with DOD-C-29501.
ESR: In accordance with DOD-C-29501.

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Resistance to soldering heat:

CAP: In accordance with DOD-C-29501.
ESR: In accordance with DOD-C-29501.

Moisture resistance:

CAP: In accordance with DOD-C-29501
ESR: In accordance with DOD-C-29501.

Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.
CAP: Within tolerance.
ESR: See table I.

Step 2 (-40°C):

CAP: Shall change not more than -40 percent from the step 1 measured value.
ESR: Shall change not more than 500 percent from the step 1 measured value.

Step 3 (+25°C):

DCL: See table I.
CAP: Shall change not more than ±20 percent from the step 1 measured value.
ESR: See table I.

Step 4 (+70°C):

DCL: Shall change not more than 500 percent from the step 1 measured value.
CAP: Shall change not more than +150 percent from the step 1 measured value.
ESR: See table I.

Step 5 (+25°C):

DCL: See table I.
CAP: Shall change not more than ±20 percent from the step 1 measured value.
ESR: See table I.

Surge voltage: In accordance with DOD-C-29501.

DCL: Shall change not more than ±20 percent.
CAP: Shall change not more than ±20 percent.
ESR: See table I.

Life: In accordance with DOD-C-29501.

DCL: Shall change not more than 200 percent of the table I value.
CAP: Shall change not more than -30 percent of the initial measured value.
ESR: Shall change not more than 120 percent of the table I value.

Resistance to solvents: Marking will be checked only.

Typical back-up current: See figure 2.

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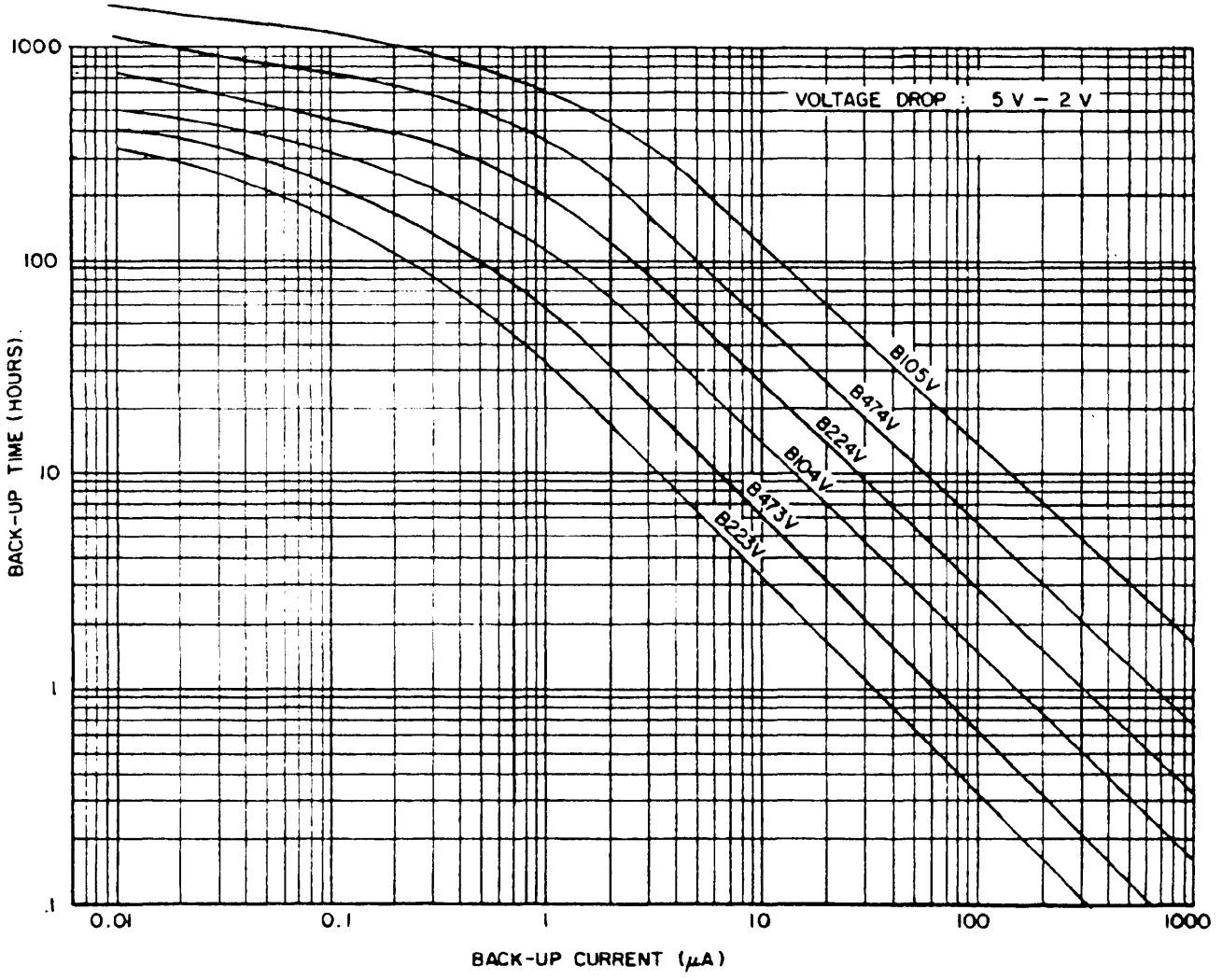


FIGURE 2. Typical back-up current at 25°C.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AV
Navy - SH
Air Force - 17, 85, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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